

1. Project Overview

This document summarizes the bugs resolved, changes made, and features restored in the Task-Manager-DevSecOps repository. The application is a full-stack task manager with a decoupled architecture featuring an Express backend, a vanilla HTML/CSS/JS frontend, and a MongoDB Atlas database.

2. Resolved Issues & Code Changes

2.1 Database Connection Fix (backend/.env)

Issue:

The backend node server failed to connect to the database, throwing a querySrv ECONNREFUSED error during startup. This was because Node's DNS resolver failed to resolve the MongoDB Atlas SRV URI structure.

Resolution:

We manually resolved the SRV and TXT DNS records for the database cluster. The MONGO_URL configuration inside 'backend/.env' was updated to use the direct replica set URI list, bypassing SRV lookup, and ensuring successful database connections.

2.2 Backend Restoration & Enable CORS (backend/server.js)

Issue:

The 'server.js' file was completely stripped of its API endpoints and schemas, which caused signup, login, and task operations to fail with CORS/preflight block and 404 errors.

Resolution:

- Installed and enabled the CORS middleware to authorize cross-origin API requests from the frontend origin.
- Restored User and Task Mongoose schemas.
- Restored '/signup' and '/login' endpoints, using bcrypt to securely hash and verify user passwords.
- Restored all '/tasks' CRUD endpoints (GET, POST, PUT, DELETE).
- Added a secure '/users-by-admin/:userId' endpoint for future administrative controls.

2.3 Frontend Fixes & Flow Improvements (frontend/)

Issues:

1. In 'dashboard.html', the 'role' variable was referenced but never declared, leading to a ReferenceError that crashed javascript execution and blocked tasks from loading.
2. On successful signup, 'signup.html' immediately redirected the user to 'dashboard.html' instead of 'index.html', which caused an authorization check bounce.

Resolutions:

- In 'dashboard.html', declared 'const role = localStorage.getItem("role") || "user";' to prevent crashes and clear the state correctly on logout.
- In 'index.html', saved the returned 'role' in 'localStorage' alongside the 'userId'.
- In 'signup.html', modified the redirect target to 'index.html' so that new users are directed to log in first.

3. Verification & Server Status

The backend node server running under nodemon has verified all changes. The startup logs show that the server is listening successfully on port 3000, and database connections are successfully verified:

```
[nodemon] starting node server.js
Server running on port 3000
MongoDB Connected Successfully
```